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Integrating Large Language Model and Digital Twins in the Context of Industry 5.0: Framework, Challenges and Opportunities

Chong Chen[a], Kuanhong Zhao[a], Jiewu Leng[b*], Chao Liu[c], Junming Fan[d], and Pai Zheng[d]

[a] *Guangdong Provincial Key Laboratory of Cyber-Physical System, Guangdong University of Technology, Guangzhou 510006, China*

[b] *State Key Laboratory of Precision Electronic Manufacturing Technology and Equipment, Guangdong University of Technology, Guangzhou 510006, China*

[c] *College of Engineering and Physical Sciences, Aston University, Birmingham B47ET, UK*

[d] *Department of Industrial and Systems Engineering, The Hong Kong Polytechnic University, Hung Hom, 999077, PR China*

[*] *Corresponding author. E-mail: jwleng@gdut.edu.cn*

Abstract

In Industry 5.0, where human ingenuity is combined with cutting-edge technologies such as artificial intelligence (AI) and robotics to revolutionize manufacturing with a focus on sustainability and human well-being, Digital Twins (DT) have become essential for real-time optimization. However, the complexity of managing DT for large-scale systems poses challenges in terms of data transmission, analytics, and advanced applications, which can be potentially addressed by Large Language Model (LLM). This research firstly performs a literature review to study the roles and functions of LLM in DT in the context of Industry 5.0. Subsequently, we propose a framework named Interactive-DT for LLM-DT integration that reveals the technical pathway for how LLM can be effectively integrated and function within DT environments. Within this framework, the roles and functionalities of LLM at the edge layer, DT layer, and service layer are elaborated upon. Finally, the identified research gaps and prospects for the integration of LLM and DT are outlined and discussed. The research outcomes of this paper highlight the potential of LLM to augment DT capabilities through improved construction and operation, enhanced cloud-edge collaboration, and sophisticated data analytics, ultimately promoting industrial practices that are both efficient and aligned with human-centric and sustainability principles in Industry 5.0.

Keywords: Large Language Models, Digital Twins, Industry 5.0, Intelligent Manufacturing.

1 Introduction

The introduction of the Industry 5.0 concept is important in the industrial development landscape. In this context, human ingenuity meets advanced technology to redefine manufacturing and production processes. [1, 2]. It follows the advent of mechanization, mass production, automation, and digitalization, underscoring a profound transformation in the way industries operate and interact with society [3]. It combines human creativity with the precision and scalability of AI, Internet-of-things (IoT), robotics, etc. The target of this combination is to increase productivity and efficiency and prioritize sustainability, worker well-being, and personalized consumer experiences [4], meeting the dynamic needs of a global economy that is increasingly focused on environmental impact and social value. One of the most important techniques of Industry 5.0 is DT, which are virtual models designed to accurately represent physical objects or processes [5, 6]. Investigating the key techniques will be helpful to improve the performance of DT.

DT is a digital replica of a physical entity that allows real-time analysis, monitoring, and control, which is an essential technology in Industry 5.0 [7, 8]. With the development of IoT, it is possible for DT to continuously connect and sync with physical assets through data collected from sensors. It can enhance operational efficiency, while reducing waste and energy consumption, contributing to the sustainability of the industrial ecosystem [9, 10]. The integration of DT with advanced techniques such as IoT and machine learning further enhances DT's capabilities, which allows for better data analytic and decision-making performance [11, 12].

In contrast to traditional methods such as simulation, which lack real-time updates, DT provides continuous feedback from the physical system. DT collects real-time data, allowing operators to see where a design may be flawed so that adjustments can be made in real-time. According to a General Electric study, a new paradigm using DT can improve operational efficiency by up to 20 percent compared to a traditional operations and maintenance paradigm without DT [13].

Thus, DT is not just a technological advancement but a foundational element that drives the innovation and adaptability required in Industry 5.0 [3]. While DT has improved the industrial landscape, several challenges including the construction and operation of DT, the communication between edge and cloud systems, and selecting AI algorithms for data analytics, have impeded its further development. The challenges limit the widespread adoption and effectiveness of DT in addressing the dynamic and diverse needs of Industry 5.0.

The introduction of LLM into this context provides a potential solution to improve the performance of DT in Industry 5.0. LLM, due to its advanced natural language processing (NLP) capabilities, can bridge the gap between the technicalities of DT and the diverse needs of human operators and decision-makers [14, 15]. Compared to conventional deep learning, LLM takes advantage of the larger model size. It shows great benefits in processing unstructured text data. While conventional deep learning models usually require rigorous preprocessing and feature extraction of the input data, and the process is complicated by the need to transform natural language into numerical features in many cases when processing textual data. Additionally, LLM can be helpful in the natural language interaction between human operators and DT, which is beneficial for more effective troubleshooting and maintenance. It also can be used in generating reports and recommendations based on real-time data, which can help operators make suitable decisions. The development of LLM represents a new milestone in AI. It was proposed to understand and generate content based on large amounts of data [16]. Deep learning techniques and the enlarged computational resources have driven the development of LLM. The ability of LLM to perform various language tasks ranging from translation and summarization to complex problem-solving and creative writing opens up new avenues for applications in manufacturing. [17, 18]. By facilitating more intuitive and accessible interfaces, LLM can democratize the use of DT, which can make them more user-friendly and adaptable across different industrial settings [19]. Furthermore, LLM can improve the predictive analytics abilities of DT by extracting, processing, and interpreting unstructured data sources, thereby improving the accuracy and relevance of insights generated. Industry 5.0 is a highly relevant context for LLM and DT, since it highlights the integration of human ingenuity with advanced technologies like LLM. By introducing LLM into DT, an interactive interface can be built that

can better understand and respond to human needs, thereby enhancing the human-centric nature of manufacturing processes.

However, the integration of DT with LLM presents a series of challenges that should be addressed to harness their combined potential in Industry 5.0. Firstly, the investigations of DT and LLM are limited. It is not clear how LLM can serve DT to leverage its performance in Industry 5.0. Secondly, there are several barriers to integrating LLM and DT in industrial settings, which include addressing the interoperability issues between different systems, managing the large amounts of generated data, which ensures the integration process is both secure and compliant with industry standards, and so on. Thirdly, the insights into how this integration can align with and contribute to the goals of Industry 5.0 need to be further explored. It requires an in-depth look at how the collaboration between DT and LLM can support industries in achieving greater sustainability, resilience, and profitability in different sectors of the industry. Therefore, this paper investigates the integration of LLM and DT in Industry 5.0 to address the challenges above. The contributions of this research are three-fold: (1) The recent advance of LLM and DT in Industry 5.0 is critically reviewed to study the role and function of LLM in DT; (2) A framework named Interactive-DT for integrating LLM and DT in Industry 5.0 is proposed; (3) The research challenges and the potential research challenges in DT-LLM are presented and discussed.

2 Literature Review

In this section, a comprehensive search of the main databases including Springer, Scopus, and IEEE, was conducted to reveal the publications which relevant to DT and/ or LLM. Among the collected papers, 25 representative publications were selected, which are detailed in Table 1.

Table 1 Classification of the Relevant and Recent Studies.

Reference	Data Type	Classification	Role of LLM	Research Field	Task
Grieves et al. [20]	Time Series Data & Setpoint Data	DT	/	Aerospace and astronautics	Monitoring and optimization
Qi et al. [21]	Time Series Data & Text Data & Setpoint Data	DT	/	Smart Manufacturing	Predictive maintenance
Wang et al. [22]	Time Series Data & Text Data & Setpoint Data	DT	/	Railway station operations	Operations management
Delgado et al. [23]	Text Data & Setpoint Data	DT	/	Built environment	Architecture, Engineering, Construction, and Operations
Mehdi et al. [24]	Time Series Data & Text Data & Setpoint Data	DT	/	Construction safety management	Optimizing fire safety measures
Liu et al. [25]	Time Series Data & Setpoint Data	DT	/	Automated flow-shop manufacturing system	Rapid individualized design of manufacturing systems
Leng et al. [26]	Time Series Data & Text Data & Setpoint Data	DT	/	Smart manufacturing workshops	Enhancing management and operation via digital twin-driven cyber-physical system
Liu et al. [27]	Time Series Data & Setpoint Data	DT	/	Production resources for shop floors	Development of a multi-scale, multi-level DT modeling method for production resource planning and simulation
Liu et al. [28]	Time Series Data & Setpoint Data	DT	/	Real-time tool condition monitoring in machining	Developing an anomaly detection framework
Ren et al. [29]	Time Series Data & Text Data & Setpoint Data	DT	/	Roll forming equipment	Development and application of a six-dimensional DT model
Magyar et al. [30]	Time Series Data & Setpoint Data	DT	/	Welding robot operations	Enhancing real-time data visualization and preparing for DT integration
Xie et al. [31]	Time Series Data & Setpoint Data	DT	/	General manufacturing systems	Enabling more general representation in DT
Huang et al. [32]	Time Series Data & Setpoint Data	DT	/	Optimal reconfiguration policy development	Smart manufacturing systems
Dou et al. [33]	Time Series Data & Setpoint Data	DT	/	Single-leg quadruped robots	Developing a virtual-real interaction system for real-time monitoring and control optimization

Chen et al. [34]	Time Series Data & Setpoint Data	DT	/	Additive manufacturing processes	Developing a service-oriented framework for AM DT
Xia et al. [19]	Text Data	LLM	/	Robotic control	Developing LLM for robotic control code generation
Zhou et al. [35]	Text Data	LLM	/	Aerospace product manufacturing	Cause analysis of quality problems
Wang et al. [36]	Text Data & Point Cloud Data	LLM	/	Human-robot collaboration	Developing a visual language model for robot navigation
Gkournelos et al. [37]	Text Data & Real-time Data	LLM	Robot behavior planning for HRC	Human-robot collaboration	Generating robot behavior sequence
Sun et al. [38]	Time Series Data & Text Data & Setpoint Data	DT-LLM	Automating parameterization and decision-making	Global temporal feature learning	Temporal Feature Learning in DT
Xia et al. [39]	Time Series Data & Text Data & Setpoint Data	DT-LLM	Improving search and parameter setting efficiency	Multi-agent systems in DT	Automating simulation parameterization
Šturm et al .[40]	Time Series Data & Text Data & Setpoint Data	DT-LLM	Improving query understanding and system transparency	Cognitive DT	Enhancing interaction with cognitive DT
Cui et al. [41]	Time Series Data & Text Data & Setpoint Data	DT-LLM	Optimizing real-time data interactions	Human-digital twin interaction in autonomous vehicles	Improving teaming dynamics
Hong et al. [42]	Time Series Data & Text Data & Setpoint Data	DT-LLM	Enhancing security and real-time analysis	Digital twin networks	Secure networking frameworks with DT
Lukaj et al. [43]	Time Series Data & Text Data & Setpoint Data	DT-LLM	Natural language processing for user interaction	Metaverse	Enhancing user interaction via speech recognition

The table demonstrates that time series data, setpoint data, and text data are the most frequently used across various domains. DT, for the most part, is utilized for monitoring, predictive maintenance, and optimization in sectors like aerospace, manufacturing, and construction. Meanwhile, LLM is more focused on tasks like robotic control, behavioral planning, and human-robot collaboration—demonstrating their strength in complex, context-specific decision-making.

The classification results provide a better scope of literature review, aimed at exploring the opportunities and challenges of combining LLM with DT in the context of Industry 5.0. The upcoming review will dig into how LLM can boost the capabilities of DT, enhancing data analysis, decision-making, and user interaction—ultimately contributing to smarter, more sustainable industrial practices.

2.1 Latest Advance of LLM and its Application in the Industry

In recent years, with the increasing growth of data and improvements in computational power, LLM has made significant progress[44]. These models, such as GPT and its variants, have broad applications in NLP tasks, which demonstrate remarkable abilities in understanding, generating, and interacting with natural language. This section provides an overview of the architectural innovations and evolving training strategies that define modern LLM like GPT and BERT [45]. LLM has made great progress in various general domains, demonstrating strong capabilities in reading comprehension, text summarization [45], text generation [46], etc. The application of LLM has brought significant advancements in various sectors, such as telecommunications [47], public planning [48], construction [23], and industry.

In industrial applications, Al and industrial convergence are showing strong industry growth. The industrial AI market size is expected to grow to $89.53 billion by 2032 [49]. It can improve the interaction, generation, and R&D innovation ability [50], and bring new power for industrial upgrading. However, the application of LLM in industrial domains is still in the beginning stage

[51]. The current study highlights a number of challenges that are specific to this vertical. One of the challenges is that these large models tend to hallucinate and produce inaccurate or meaningless answers. [52]. LLM has extensive knowledge but low information accuracy, which limits the application in industry and direct decision-making scenarios. LLM can achieve an answer accuracy of 80% [53], but in industrial scenarios, accuracy needs to be 99%+ or even 100%. Additionally, LLM often struggles with real-time responses due to slower processing speeds, which do not meet the stringent requirements of industrial settings.

Fine-tuning generalized LLM, which can retrain the existing models with industry-specific datasets. It bridges the gap between generalized pre-trained models and application-specific requirements to make LLM more relevant to real-world production needs, and training specialized LLM is an effective way to improve the hallucination problem [44, 54]. Widely used techniques are Proximal Policy Optimization [55] and Direct Policy Optimization [56]. LLM can act as a cognitive assistant to guide production processes, Freie et al. [57] utilized LLM to support cognitive assistants, which can leverage the knowledge-sharing level of the manufacturing process. They applied GPT-3.5 in the textile and detergent industries to answer questions, provide task guidance, and solve machine problems. Further, Samuel Kernan Freire et al. [58] leveraged the in-context learning abilities of LLM to provide them with information that is pertinent to the user's query. Ezukwoke et al. [59] introduced the Big GCVAE model, which leverages LLM for failure root cause analysis in the semiconductor industry. The methodology aims to optimize the representation, reduce reconstruction loss, and improve decoupling in the potential space. Fan et al. [60] proposed an industrial-grade federated learning framework FATE-LLM. The authors used parameter-efficient fine-tuning methods (e.g., LoRA [61] and P-Tuning-v2 [62]) to facilitate LLM training, which significantly reduces the communication cost and local model adaptation costs required during the training process.

Knowledge Graphs (KGs) support persistent memory and context association. Integrating it with LLM can enhance the practicality and reliability in the manufacturing environment [63]. KGs provide structured information that can help LLM verify facts and reduce hallucinations. Pan et al. [64] incorporated KGs into the pre-training and inference phases of LLM, which

effectively boosted the accuracy and reliability of the models. In the pre-training phase, KGs provide structured knowledge that lowers the probability of model hallucination and improves the LLM's comprehension of domain-specific knowledge. In the inference phase, the real-time knowledge retrieval function of KGs helps LLM generate more accurate and relevant responses [65, 66]. The cooperation between LLM and KGs is reflected in the enhancement of knowledge representation and inference capabilities. Researchers have designed a unified framework that enables LLMs to dynamically interact with KGs for efficient knowledge retrieval and reasoning by integrating the strengths of both [64]. Such fusion can improve the performance of LLMs in complex tasks and leverage their interpretability, making them more trustworthy in high-risk applications [64].

Embodied AI refers to intelligence that has a body and supports physical interaction, enabling an agent to learn by interacting with its environment.[67]. Embodied intelligent robots refer to robots that can interact with the environment to perceive, self-help planning, decision-making, action, and execution of tasks, just like humans. Fan et al. [68] illustrated LLM agents applying zero-shot and few-shot learning techniques to derive the tasks and associated process parameters in manufacturing scenarios from language descriptions, which makes industrial robots independently plan, make decisions, and carry out tasks in manufacturing settings. Driess et al. [69] proposed PaLM-E, a multimodal LLM that realizes the connection between words and perceptions by embedding visual, linguistic, and sensor data into a pre-trained LLM, enhancing its real-world reasoning and decision-making capabilities to handle a variety of embodied reasoning tasks from different observation modalities.

Despite these applications, several challenges persist in deploying LLM in industrial environments. Muthusamy et al. [70] discussed the limitations of LLM-based autonomous agents, which mainly focus on enterprise settings. They also highlighted issues such as the brittleness of LLM, which can lead to inconsistent outputs and the high costs associated with their deployment. The paper also highlights the need for explainable and trustworthy systems, especially given the potential for significant impacts from autonomous actions taken by these agents. The lack of robust metrics and benchmarks to evaluate LLM's performance in complex

tasks further complicates their deployment in enterprise environments.

2.2 Recent Studies of Digital Twins in Industry 5.0

DT is a technology that simulates, analyzes, and optimizes the operational state and behavior of physical entities (e.g., devices, products, or processes) by creating their virtual copies. These virtual models are interconnected with their real-world counterparts, enabling real-time data analysis, predictive maintenance, and performance optimization [20, 21, 71]. A DT can be defined as a dynamic, real-time virtual representation that reflects the state and behavior of a physical entity, including its physical characteristics, processes, and environmental conditions [9, 64]. With DT, it is possible to simulate and optimize the operation of an entity in real-time, predict its future behavior, and support decision-making and problem-solving [9, 13, 22, 72]. DT mainly consists of the physical product in physical space, the virtual product in virtual space, and the interface for data and information interaction between the interface for data and information interaction between physical and virtual space [71].

Tao et al. [13] summarized the development process of DT up to 2018 and explored the application of DT in manufacturing. The development of DT can be traced back to 2003 and has gone through three main phases formation, incubation, and rapid development up to 2018. The concept of DT was first introduced by Michael Grieves in 2003 for use in product lifecycle management in teaching [13]. The DT model in this period mainly contained physical products, virtual products, and their connecting parts The development of the IoT, big data analytics, and simulation laid the foundation for further development of DT at that time. In 2011, the first journal article on DT was published, which explored the use of DT in predicting the life of aircraft structures In 2012, the National Aeronautics and Space Administration (NASA) defined DT in more detail, conceptualized DT as "the integration of a multidisciplinary, multi-scale simulation process that makes full use of physical models, sensors, operational history, and other data, which serves as a mirror image of the physical product in virtual space and reflects the full life-cycle process of the corresponding physical entity product", emphasizing its

promise in the aerospace industry [73]. Since 2011, the number of conferences and journal articles on DT has increased rapidly, demonstrating the growing academic interest in the field. The applications of DT are beginning to expand into other industries. In 2014, the first white paper on DT was published [74], marking the evolution of DT from conceptual ideas to practical applications.

Conceptualization is the first step in the product design. To generate a satisfactory design concept, the designer should consider data, information, and knowledge from various sources such as market, customers, manufacturing, and internal parties [72]. Tao et al. [75] proposed a DT approach to product design based on big data. In this study, a digital twin-driven product design (DTPD) framework was constructed. DTPD collects data about physical products and consumer interactions through sensors or derives virtual data through interviews, downloading from internet sources, or reviews of documents. Data from multiple sources is then integrated to provide users with a comprehensive view. Data purification is carried out by detecting, eliminating, and rectifying various inaccuracies within the DTPD dataset. Finally, data mining is performed to extract information from the dataset and transform it into a visual structure for further use.

DT is utilized extensively in maintenance and operation management within industrial settings. Wang et al. [76] designed a DT model for fault diagnosis of rotating machinery is proposed. The proposed model monitors the actual operating condition of the physical system, collects data on the geometry, operating state, and environmental conditions through intelligent sensors, and combines the physical model with data-driven intelligent analysis for health state analysis and maintenance decisions. To make sure the digital model can truly reflect the actual state of the real-world system, the article proposes a strategy based on parameter sensitivity analysis to update the model. Errandonea et al. [71] provided a comprehensive review of DT applications, emphasizing their role in prescriptive maintenance, design, logistics, and production processes. DT can also facilitate asset health management, reducing downtime and improving operational efficiency. A study conducted by Chen et al. [9] discussed how machine learning integrated with DT can predict equipment failures before they occur, thus preventing costly interruptions.

To address the problem that the faulty data samples are often limited, transfer learning is introduced to facilitate fault diagnosis—the use of Asset Administration Shell (AAS), to overcome the challenge of multi-source data fusion. The proactive maintenance approach is not just about preventing failures but also optimizing the manufacturing process for better productivity and reduced wear and tear on equipment.

Despite the enormous potential of DT, their use in industry also faces several challenges. Firstly, the development of a high-fidelity model that aligns with real-world requirements; secondly, the construction of DT models frequently necessitates the combination of interdisciplinary expertise; and thirdly, DT models often necessitate the gathering of sensor data from an array of sources, posing the question of how to incorporate these data, which may differ in format and frequency of acquisition, into the digital twin model. Fourthly, the challenge of facilitating user interaction with DT. The fusion of LLM with DT could represent a pathway to address these issues. The integration of LLM with DT presents substantial opportunities for innovation and technical challenges. It has the potential to augment data processing and decision-making capabilities and to provide more intelligent and automated solutions.

The basic components of the digital twin are the physical product, the digital model, and the connection between the physical and virtual worlds [13]. In the early stages of product design (i.e., conceptual design and idea generation), there are challenges in applying DT to product design because there are no physical prototypes available to collect real-time data (e.g., user experience, impact on environmental changes, etc.) [72]. Using DT to assist and optimize the idea-generation process can be difficult for new ideas or concepts. LLM can extract information from a large amount of literature, market reports, and user feedback to provide designers with a rich source of ideas [77]. By analyzing this data, LLM can generate potential design concepts and ideas, help designers generate virtual models based on existing data and knowledge at an early stage for initial simulation and performance prediction, and help designers validate and optimize without actual physical prototypes.

Furthermore, Šturm et al. [40] discussed integrating LLM agents at a national level for cognitive

DT, providing intuitive conversational interfaces that allow users to query and interact naturally. It can improve user experience and offer transparent and explainable decision support, aiding decision-makers in making more informed choices. By combining multimodal data from sensors, web pages, and human inputs, LLM agents can integrate and analyze this information, providing real-time, dynamic system performance, which is essential for fields such as policy-making, urban planning, and resource management, helping stakeholders understand complex systems and predict future behaviors.

In 2021, the European Union proposed the concept of "Industry 5.0", whose core elements are human-centric, sustainability, and resilience. Industry 4.0 prioritizes process automation to minimize human intervention in the manufacturing process, while Industry 5.0 focuses on collaboration between humans and machines, especially combining human creativity and innovation with machine intelligence and efficiency [78]. It emphasizes the fusion of artificial intelligence and human intelligence. DT can improve the accuracy of production planning and quality control, while it also revolutionizes production efficiency [3].

Industry 5.0 treats humans as the core of the production process. Human creativity, flexibility, and decision-making ability are irreplaceable, and technological advances should be combined with human potential. Pang et al. [79] highlighted the importance of manual operations in the assembly of small batches and customized products, which indicates that despite the continuous development of automation technology, human flexibility and adaptability still cannot be replaced in specific scenarios. Potential errors in the assembly process can be prevented through digital twin technology. Ignatius et al. [80] applied DT in logistics systems to improve productivity while ensuring fairness and prioritizing worker needs in the collaboration between humans and automated equipment. Caputo et al. [81] integrated ergonomics in workplace design through digital twin technology, which allows designers to simulate work tasks in virtual environments in order to identify and correct ergonomic problems earlier, improving productivity and reducing the risk of design errors and worker injuries.

The integration of LLM with DT is also a transformative approach in Industry 5.0, aiming to

enhance the capabilities and efficiency of DT technology through advanced natural language processing and cognitive abilities. LLM, such as ChatGPT, Qwen, Claude, and GLM, have demonstrated advancements in understanding and generating content in different formats, which can bridge the complex technical data and human-readable formats, thereby making DT more accessible and user-friendly in Industry 5.0, embodying a human-centered approach. The integration can be beneficial in industrial settings where the data generated by DT needs to be explained and acted upon by human operators and decision-makers. With their powerful NLP capabilities, LLM can understand and generate complex language information [40]. For example, Xia et al. [39] introduced a multi-agent system framework that uses LLM to automate the parameterization process of DT models. The framework has the key features of agents for observation, reasoning, decision-making, and summarization, which are beneficial to dynamic interaction and intelligent control over complex simulation scenarios. It autonomously explores parameter settings, optimizes simulation model performance, reduces the need for human intervention, and enhances operational efficiency.

The meaning of sustainability in Industry 5.0 is two-fold: firstly, building a modern, resource-efficient, and sustainable industry and transitioning to a circular economy, with a focus on environmental protection in the production process; and secondly, continuing to provide skills training for employees so that workers remain competitive in this fast-moving technological age. Zhang et al. [82] proposed a framework for big data-driven product lifecycle management to support enterprises in analyzing and reducing waste generation and emissions throughout the lifecycle. Big data-driven product lifecycle management can also help companies provide longer maintenance services, extend product lifespan, reduce the need for new products, and thus reduce the consumption of natural resources. Rojek et al. [83] incorporated the 6R approach (Reduce, Reuse, Recycle, Recover, Redesign, Remanufacture), which is central to sustainability in Industry 5.0. By using digital twins to simulate and optimize product lifecycles, manufacturers can achieve higher resource efficiency and reduce their environmental impact.

Resilience refers to the need for companies to adapt quickly to changes in the marketplace and adjust production in time to stay ahead of the curve. In Industry 5.0, the primary focus of

business strategy will no longer be on growth, profitability, and efficiency, but rather on building ‘non-vulnerable’ organizations that can anticipate, respond, and learn from any crisis in a timely and systematic manner, thus ensuring stable and sustainable performance [1]. Hong et al. [42] developed an "LLM-Twin" framework, an integration of LLM with DT in a networked environment, aimed at enhancing data processing and security for next-generation communication systems. The proposed model employs a dual-structure approach, where localized 'mini' models handle specific, edge-level tasks and a centralized 'giant' LLM processes large-scale data, significantly optimizing computational efficiency and reducing network load. The framework also introduces semantic secure communication techniques, including homomorphic encryption, to ensure data integrity and confidentiality during transmission. Despite its advancements, the LLM-Twin framework's reliance on complex encryption and the high computational power of LLM might limit its applicability in smaller-scale or resource-constrained environments.

From existing literature, it is evident that LLM and DT have broad applications in Industry 5.0. However, it is not clear how to integrate LLM and DT to leverage Industry 5.0. When deployed individually, LLMs may suffer from reasoning errors due to incomplete or limited information sources and DT may suffer from a lack of multi-modal data analytics tools. Integrating DT with LLM can create a complementary system, which takes advantage of both techniques. DT can simulate and validate the recommendations provided by LLMs in its virtual environment, which can help identify and correct errors and avoid the direct implementation of flawed decisions. When DT detects deviations in the reasoning outcomes of LLMs, the simulation feedback can be used as training data to drive iterative optimization of the LLM. Leveraging the natural language processing and reasoning capabilities of LLMs, DT can extend its functionalities, such as generating user-friendly reports, predicting unknown failure patterns, and optimizing operational workflows in complex systems. Such an integration can address the limitations of standalone deployment and improve the efficiency and accuracy of industrial processes.

3 Interactive-DT: A Framework for Integrating LLM and Digital Twins in Industry 5.0

Based on the existing literature, we propose a research framework named Interactive-DT to integrate LLM and DT. Interactive-DT not only enhances the interactions among different components within DT, but also enhances the interactions between the DT and human operators, specifically in Industry 5.0. The research framework is composed of four interconnected layers that work together to optimize operations, maintenance, and control strategies. By leveraging the advanced natural language processing capabilities of LLM, Interactive-DT leverages the interactivity and responsiveness of the DT, making it a powerful tool for the collaboration of humans and machines in Industry 5.0.

The architecture of Interactive-DT is shown in Figure 1. The Asset Layer, responsible for data acquisition and preliminary processing, consists of sensors, actuators, machines, RFID tags, and edge devices. The Edge Layer, utilizing edge computing, houses local data storage, wireless networks, and computing nodes, where LLM facilitates contextual data annotation, decision support, and interaction between DT agents. The Digital Twin Layer acts as a high-fidelity link between physical and virtual systems, which can be beneficial for seamless data fusion across design, manufacturing, logistics, and maintenance. It integrates physical data through digital mapping and updates, with LLM functioning as intelligent agents, leveraging knowledge graphs for deep learning and data processing. The Service Layer encompasses product life cycle activities, such as design, manufacturing, maintenance, and recycling, and enhances customer engagement and materials analysis through cognitive design, dynamic process optimization, interactive troubleshooting, and supply chain coordination. Standardized interfaces between humans and systems, between layers, and within layers allow for seamless collaboration and scalability.

The Asset Layer includes sensors, actuators, machines, RFID tags, and edge devices. These devices generate multiple data types, including sensor data and operational logs. Sensor data covers various physical parameters like temperature, pressure, and vibration, with different

sampling frequencies. Ensuring temporal consistency is important for accurate analysis. The operation log data captures the equipment's operational status, fault information, and maintenance records. These data sources often vary in format, structure, and quality. To solve these problems, an Edge Layer is needed to process this data. Data from the Asset Layer is transferred to the Edge layer through the data interface using MQTT or CoAP protocols to ensure real-time performance.

Subsequently, the Edge Layer processes a large number of data from the Asset Layer through edge computing. To ensure timely data acquisition and transmission, multiple edge nodes are often deployed. Introduce an edge computing scheduling system (e.g., Kubernetes on Edge) that dynamically allocates compute and storage resources based on load. Edge computing nodes perform preliminary processing on the aggregated sensor data, such as filtering, compression, and feature extraction. Sensor data with different formats, qualities, and frequencies are harmonized into standard formats (e.g., JSON, Protobuf) with timestamps at this stage to facilitate later analysis. At the same time, the computational power of the edge nodes is used to perform some critical tasks, such as Fast Fourier Transform (FFT) analysis of vibration data or extraction of device state features from images. Some simple fault detection can be done at the edge node (e.g., whether the temperature is out of range). Through these processes, the data becomes smaller and better characterized, reducing the burden of subsequent transmission and analysis. The processed data is subsequently transferred to the cloud for more advanced analytics and DT modeling. The latency and bandwidth requirements for data transmission are reduced. At this layer, LLM can be used for the annotation of contextual data, labeling relevant operation logs into the time series data to help understand and categorize the data. They provide real-time analytics for local decision-making and determine the features, and data in specific periods that require to be uploaded to the cloud, thereby improving system responsiveness and reducing computational load on the cloud. The next step in the transmission of the aligned timestamped Unified Format data is to transfer it to the Digital twins Layer, a process that may involve propagation over the public network and requires a combination of low latency, reliability, and security. Data compression techniques can be introduced to reduce the size of the data, and encryption algorithms such as AES can be introduced to protect the data, but at

the same time, the additional processing time introduced by the steps of decompression, encryption and decryption of the data should be considered. If necessary, technologies such as VPN and Zero Trust are used to avoid unauthorized access.

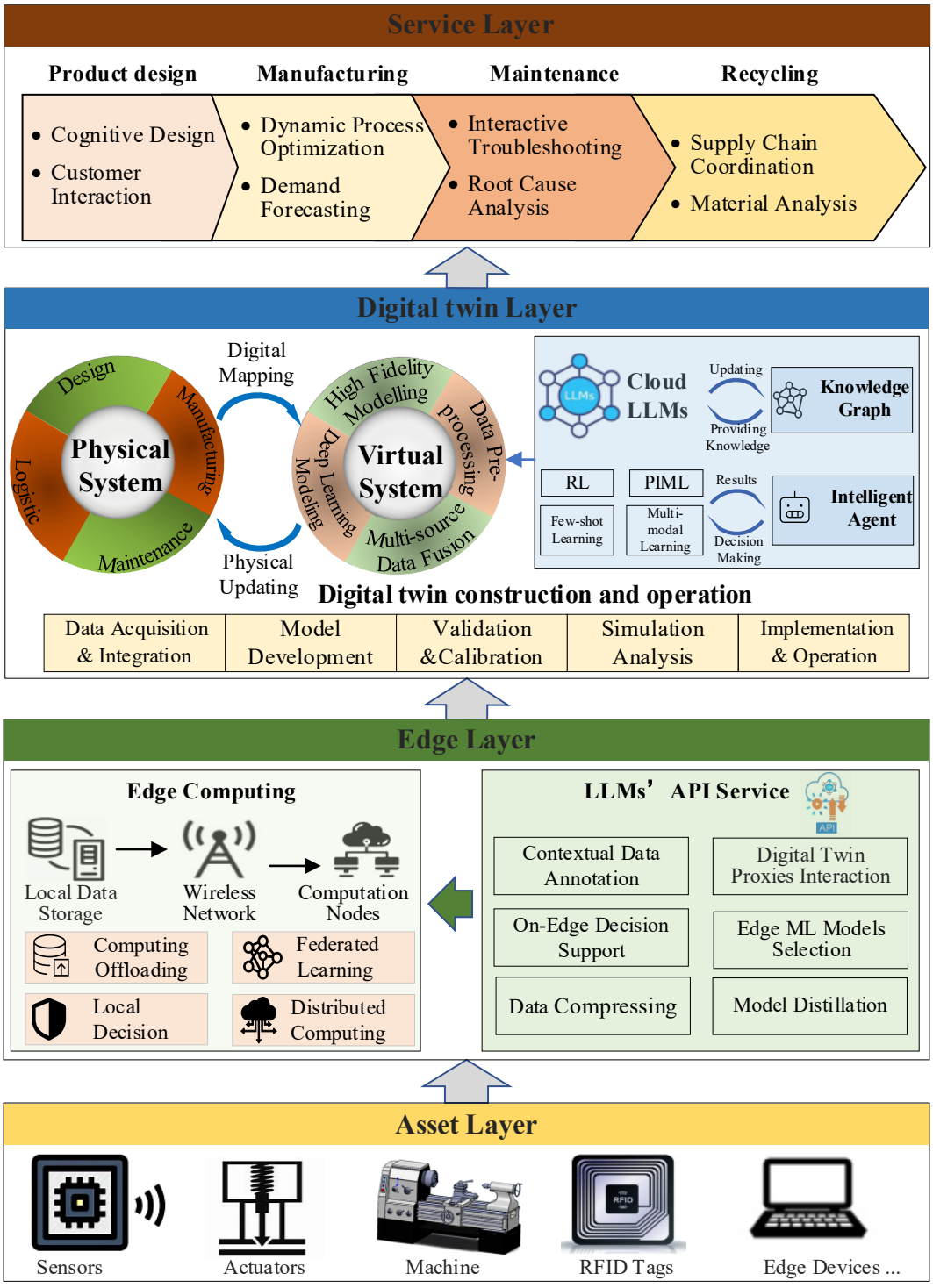


Figure 1. Interactive-DT framework for integrating LLM and DT in Industry 5.0.

The data from the Edge Layer is then sent to the Digital Twin Layer, which is a virtual model of a physical object that enables monitoring and optimization of the entire chain of physical systems from conceptual design to equipment manufacturing and maintenance through high-fidelity modeling and multi-source data fusion. The digital twin can monitor the state of the physical system in real-time, and collect and analyze data from the edge layer. By analyzing historical data and applying machine learning algorithms, DT can predict the future state of the system, which can bring tangible benefits for operations and maintenance. DT can test and validate in virtual environments before being implemented in physical systems, reducing risk and cost. LLM provides a more comprehensive and in-depth system understanding by building and maintaining knowledge graphs that correlate data from DT. LLM intelligently processes and analyzes multi-source data, improving accuracy and validity. They support decision-making and task execution through high-fidelity modeling of DT and multi-source data fusion They can understand and generate natural language and support human-computer interaction, enabling operators to control and manage the system more intuitively. The collaborative interface between LLM and DT requires the integration of various functionalities, including data sharing, tasking, real-time feedback, and interpretation of results. The structured data provided by DT can be converted from time series data or logs to text using a specific data preprocessing tool, such as Pandas or NumPy. Logs can be converted to text, while image data can be analyzed by LLM using edge processing tools (e.g. OpenCV). LLM can also export structured data for direct use by DT, and collaboration between LLM and DT is described in Section 3.3.

Finally, in the Services Layer, LLM-DT plays a pivotal role. The collaborative interface between LLM and DT requires integration in terms of data sharing, task allocation, real-time feedback, and result interpretation. The LLM outputs structured data that can be directly utilized by the DT. A detailed explanation of the collaboration between LLM and DT will be presented in Section 3.3. In the product design stage, LLM-DT facilitates innovative designs through cognitive and detailed design [77], enhances customer engagement, and provides personalized suggestions and feedback. During the manufacturing process, LLM-DT can help analyze data collected in real-time, optimizing dynamic workflows, boosting productivity, and accurately

predicting demand, thereby optimizing production schedules. In the maintenance stage, LLM-DT can diagnose issues by analyzing sensor data and operation logs, offering interactive troubleshooting solutions, and conducting root cause analysis to identify equipment failures and propose preventive measures. In the recycling phase, LLM-DT can help optimize supply chains, allocate resources efficiently, and improve logistics, while analyzing material composition and usage to suggest optimal recycling and reuse strategies, thereby promoting a circular economy. These capabilities significantly enhance product lifecycle management's efficiency and financial returns, improve customer experience, reduce operational costs, and foster sustainable development.

3.1 LLM for Digital Twin Construction and Operation

Aspects of construction and operation are the most crucial aspects of DT, which make it possible to correlate and optimize the physical systems using advanced data processing modeling and simulation techniques. The technological method of realization of LLM for DT building and functioning is illustrated in Figure 3. This part describes the step-by-step procedure of the manufacturing and use of the digital twin, showing the most essential tasks that LLM performs for each step of the transition from data collection to application and communication with users so that an integrated and efficient process is realized.

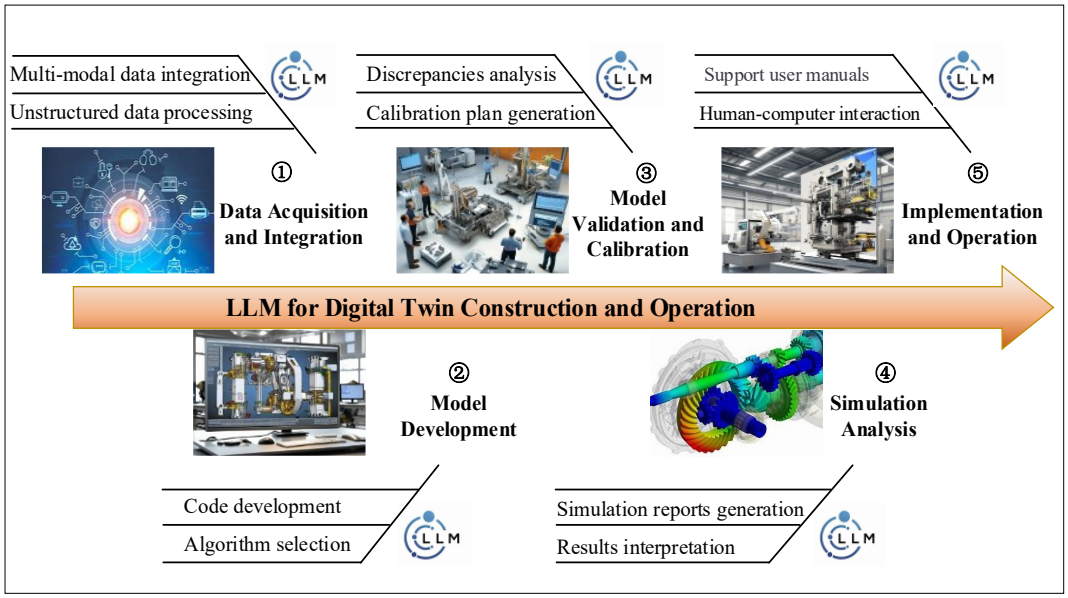


Figure 2. LLM for digital twin construction and operation.

Stage 1 Data Acquisition and Integration:

As shown in Figure 1, data is gathered from different sources, such as sensors, IoT devices, and other equipment within the Asset Layer in the manufacturing setting. The multi-source data makes sure DT reflects the correct state of physical systems. However, these data often have different formats and require further processing before they can be used for DT modeling. LLM is good at interpreting data in different formats, such as textual reports and log data. Then transforms and integrates these data and structures to create a unified database. LLM also can combine with Retrieval-Augmented Generation (RAG) [84], which can query useful information from external knowledge bases (e.g., databases or large text corpora). RAG can enhance the ability of weak model LLMs to deal with tasks that require task-specific information, especially when up-to-date knowledge or specialized knowledge is required. In the use of LLM, the system makes sure that processes for integrating various types of data and metadata are carried out optimally and the data types included in the DT are created: structured, semi-structured, or unstructured. The precision and completeness of the gathered data can be enhanced during the process of information gathering. Once the data is acquired and integrated, it undergoes pre-processing to ensure consistency and accuracy. This step involves cleaning and normalizing the data and handling missing values and outliers. LLM plays an important role in this process by generating explanations or summaries of the preprocessing steps, aiding

in the understanding of the data-cleaning process. LLM provides natural language descriptions of data transformations and quality checks, making the preprocessing workflow more transparent and easier to follow. Through the data descriptions, LLM is used to maintain data integrity, ensuring that the data used for DT modeling is reliable and high in modeling quality.

Stage 2 Model Development: Model Development is a key step in DT. It determines the goals and intended use of the DT based on whether it is used for predictive maintenance, which can optimize operations, or simulate different scenarios. The appropriate model type is then selected with the consideration of the goals and data characteristics, such as physical models, statistical models, machine learning models, etc. To develop a mathematically or computationally accurate representation of a physical system, it is important to choose suitable algorithms and techniques that can effectively simulate its behavior. LLM can exploit their knowledge of data characteristics and problem domains to propose model architectures and recommend algorithms. After that, the model structure needs to be defined to determine the relationships and interactions between the model's inputs, outputs, and internal state variables and variables, and LLM can provide specific model architecture suggestions, such as which neural network structure to use. During model parameterization, LLM can automatically perform hyper-parameter tuning to help find the optimal combination of model parameters and improve model performance [85]. Use techniques such as Bayesian optimization [86], grid search, or stochastic search to optimize model parameters. LLM can automatically generate evaluation reports that summarize the model's performance, advantages, and limitations, thereby offering reports to help users understand the model's performance and potential problems.

Stage 3 Model Validation and Calibration: After the digital twin model is constructed, validation and calibration are needed. Validation and calibration assess the model performance on different evaluation datasets, which can evaluate the model’s capability to reflect the behavior of a real system. Validation and calibration are essential in identifying model biases and instabilities and improving the reliability and robustness of models. The validation and calibration process improves the generalization ability of the model by ensuring that the model can perform well on training data as well as the unseen data. In the process of model validation

and calibration, the dataset first needs to be divided into training, validation, and test sets, and LLM can automate this process by providing best practice recommendations for data segmentation. In this scenario, time-series LLM that can analyze the model error can be a useful tool. LLM can also provide explanations and application scenarios for each metric based on the specific model and task, recommending the appropriate metrics to be evaluated (e.g., accuracy, precision, mean square error, etc.). During the model validation and evaluation process, LLM can test the model on the validation set, calculate the evaluation metrics, and generate an evaluation report.

Stage 4 Simulation Analysis: Simulation analysis is an important phase in developing and operating DT. It contains the creation of detailed and dynamic models that replicate the status of physical systems under various conditions. The simulations can be used for understanding system operations, identifying potential issues, and testing different scenarios without relying on physical experiments. The historical simulated data also can be used for the fine-tuning of LLM. It makes it possible to evaluate and predict system behaviour aiding in process improvement and better decision-making. LLM can fully assist the simulation analysis process, especially when it comes to simulation report generation and result analysis[87]. LLM performs the documentation process automatically via the generation of comprehensive reports on the simulation setup and an overview of the processes and results with explanations, illustrations, and recommendations. This ability can translate complex technical details into clear and concise language, making simulation findings accessible to diverse audiences across teams including non-technical users. It saves time in communication and ensures consistency and thoroughness in reporting. Furthermore, LLM does well in the task of evaluating simulation results by analyzing outcomes to learn the patterns, anomalies, and insights that might not be obvious. Leveraging their deep learning capabilities, LLM highlights significant trends and correlations, providing a rich understanding of the simulation data. The analysis helps engineers and decision-makers grasp the implications of the simulation outcomes, leading to more informed and effective decisions. LLM further enhances DT's ability by presenting insights in natural language, making complex data more understandable and actionable.

Stage 5 Implementation and Operation: The deployment of DT is the next stage, enabling its use to monitor, control, and optimize the physical system.

This step involves integrating the DT with the existing systems, setting up real-time data streams, and ensuring seamless operation. Before the system goes live, LLM can provide direction for deployment while also creating scripts that can make the process easier while ensuring security. LLM facilitates the selection of deployment options which include cloud, on-premises, and edge computing which is important for efficiency and cost. Also, LLM is capable of automating the capturing, processing, and storage tasks so that the entire operational data is guaranteed accuracy while still being readily accessible for analysis. They offer advanced algorithms to identify trends, patterns, and anomalies in the data, supporting ongoing optimization and improvement of the DT. LLM can generate detailed visualization reports, making it easier for users to understand the DT's performance and make informed decisions. Furthermore, LLM creates comprehensive documentation, including user manuals, technical guides, and operational procedures, ensuring users have all the information needed to effectively operate the DT. In this way, LLM helps users quickly master how to interact with and leverage the digital twin's capabilities. This way, the use of LLM provides useful strategies on how to enhance the performance of the DT in order to improve the efficiency of the system.

3.2 LLM for Cloud-edge Collaboration in Digital Twins

In industrial production, the large amount of data places high demands on computing power. At the same time, there are many decisions must be made on time, which puts a demand on the responsiveness and accuracy of LLM and DT. Deploying extensive computing power in a factory to meet peak demand for computing power at a specific demand result in increased upfront investment and resource inefficiency. Cloud-edge Collaboration can solve these problems by leveraging the computational efficiency of the cloud with the low-latency processing of edge devices. It can improve the responsiveness and accuracy of DT and reduce computational costs, especially in dynamic and large-scale industrial environments. It uses low latency, high bandwidth, and localized processing power of edge devices. By distributing the

computational load and data processing tasks between the cloud and edge, DT can enhance operational efficiency, reducing latency, and improving data handling capabilities.

LLM can be a helpful tool to improve the performance of cloud-edge collaboration in DT. Figure 4 shows the role and function of LLM in cloud-edge collaboration in DT.

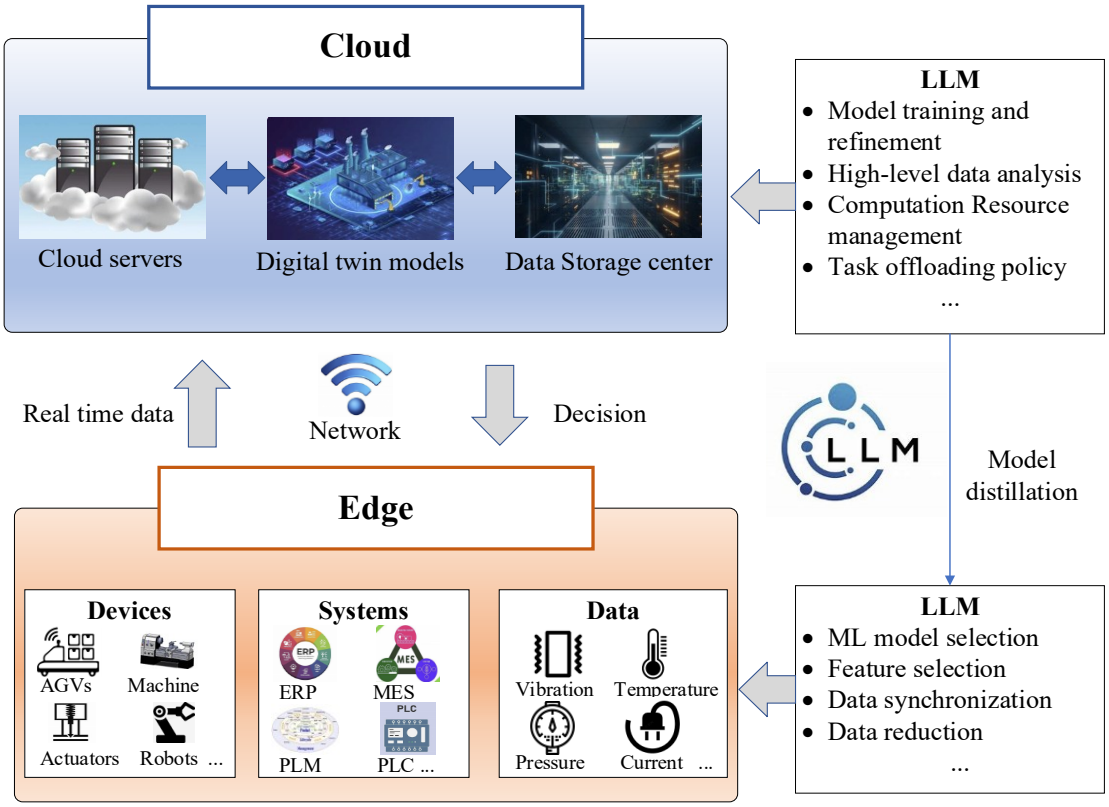


Figure 3. LLM for cloud-edge collaboration in DT.

3.2.1 LLM for Edge Computing

Industry 5.0 aims to tightly integrate human intelligence with AI [88], enhancing productivity and efficiency through highly customized and adaptable production methods. At its core, DT serves as a driving force, which according to the creation of virtual copies of physical assets helps in real-time monitoring and optimization of their performance. However, the deployment of DT necessitates the efficient handling of substantial amounts of data collected in real-time, primarily sourced from sensors, IoT devices, and other systems. The multi-source data is not only extensive but also demands extremely rapid and precise processing. Consequently, edge computing has become an indispensable component of DT in Industry 5.0.

Edge nodes have limited computational power and should be used to reduce the size and complexity of the ML model without compromising performance through techniques such as model distillation, pruning, and quantization can be used to reduce the size and complexity of the ML model [89]. In the optimization process of ML models, LLM can help this process. In cloud-edge collaboration scenarios, edge nodes often perform a large amount of data collection and processing. LLM can be used to automate pre-processing tasks such as data annotation, data filtering, data cleaning, etc. to help ML algorithms train better. LLM can be designed to automatically trim unnecessary parameters or nodes to reduce the model's computational and storage requirements, thus optimizing the performance of ML algorithms on edge nodes.

Edge computing relies on a distributed architecture, ensuring that even if some edge nodes fail, other nodes can continue to operate. Edge computing also allows model updates and maintenance to be performed locally [42], reducing dependence on servers. This is where resilience comes into play in Industry 5.0. The multi-source data collected at the device layer often have different formats (e.g., time-series data and text data) and may be received asynchronously, LLM can be employed to design and manage efficient data synchronization mechanisms to preprocess the data, ensure data consistency among multiple edge nodes, and reduce data conflicts and inconsistencies. Processed data can then be selectively uploaded to the cloud, where it is intelligently compressed and optimized to lower data transmission demands and improve efficiency.

3.2.2 LLM for Cloud Computing

DT needs to process a large amount of real-time data and perform advanced simulations and analyses. Given the limitations of edge devices, many computations and decisions are offloaded to the cloud, which offers virtually unlimited computing power. The integration of LLM in cloud computing enhances the cloud's capabilities, supporting DT with advanced analytics and predictions through big data. DT requires massive storage of historical and real-time data, combined with the accurate analytics and predictive capabilities of big data, to be more responsive and predictive in industrial applications [21]. This is made possible by the virtually

unlimited and dynamic scalability of the cloud. LLM can analyze large datasets stored in the cloud to identify patterns, trends, and anomalies, thereby providing valuable insights into system behavior and performance. Additionally, the use of cloud resources reduces costs by limiting the need for local hardware and maintenance through an on-demand resource model. LLM can also optimize cloud resource allocation, further improving efficiency. Meanwhile, LLM can also be used for distributed model training [90, 91], leveraging the computing resources of multiple edge devices and the cloud, which enables training on large datasets and ensures model accuracy and generalization.

3.2.3 Interaction between Edge and Cloud

The interaction between edge and cloud computing is essential for leveraging the benefits of DT in Cloud-edge collaboration. LLM facilitates this interaction in several crucial ways. LLM facilitates this interaction in various ways, including optimizing resource utilization and allocation.

Computing offloading refers to resource-constrained devices migrating some or all of the resource-intensive computation from mobile devices to resource-rich nearby infrastructures to address the shortcomings of edge devices in terms of resource storage, computational performance, and energy efficiency. Dong et al. [92] proposed the LAMBO framework aims to address several challenges in traditional offloading architectures, such as heterogeneous constraints, partial perception, uncertain generalization, and lack of adaptability in dynamic environments. It demonstrates that LAMBO achieves superior performance in task offloading and resource scheduling compared to traditional deep learning methods. It is highly adaptable to diverse tasks and operates efficiently in dynamic environments.

Deploying LLM on publicly accessible infrastructure raises the likelihood of attacks, and organizations may not fully understand the robustness of the security measures in place [93]. By processing sensitive data locally, the risk of data leakage is minimized, as less data is transmitted to the cloud [95]. Federated learning enhances model performance by coordinating distributed edge devices for collaborative training without sharing raw data, thereby improving

data privacy and security. Furthermore, distributed computing capitalizes on the combined resources of edge and cloud, enabling parallel processing and efficient task execution through resource allocation.

Real-time decision-making is often crucial in industrial scenarios, which is facilitated by edge processing. By processing data locally or near the source, edge computing significantly lowers latency, thereby meeting the need for immediate data handling. However, for tasks that require both substantial computational power and low latency, the latency needs to be minimized as much as possible. The vector database-assisted cloud-edge collaborative LLM QoS Optimization (VELO) framework, employs a vector database to cache the results of some LLM requests at the edge to reduce the response time of subsequent similar requests [94]. Additionally, deploying LLM in edge nodes to preprocess data, such as filtering, compressing, and extracting features, only transmitting valuable information to the cloud, can reduce the need for multiple data transfers and allows digital twin networks to process real-time data more efficiently [42]. The combination of edge computing and LLM enhances the ability of edge nodes to make autonomous decisions based on real-time data and reduces the frequency of offloading task to the cloud, reducing latency and improving responsiveness. LLM also monitors the health of the cloud system at the edge, detecting potential failures and initiating a recovery process that reallocates resources to maintain system stability and performance.

LLM enhances operational efficacy by minimizing latency and bandwidth necessities through optimized data processing, model training, and decision-making procedures. LLM augments performance by refining models for more precise predictions and simulations. The deployment of LLM across a variety of edge devices and cloud platforms facilitates the creation of adaptable and scalable solutions. Furthermore, autonomous operations are enhanced as edge devices gain the ability to make independent decisions, thereby minimizing human intervention and accelerating response times. The integration of LLM into Cloud Edge Computing provides a robust foundation for DT operations, improving efficiency, performance, and user experience while addressing challenges related to resource constraints, data security, and model complexity.

3.3 LLM for Digital Twins' Advanced Data Analytics

Advanced analytics is defined as the process of utilising sophisticated machine learning (ML) and visualisation techniques with the objective of gaining insights into data that extend beyond the scope of traditional business intelligence. The integration of LLM with DT can facilitate advanced data analytics, thereby enhancing the capabilities and efficiency of digital twin systems. By capitalizing on the distinctive advantages of LLM, the efficacy and utility of the digital twin can be enhanced. As illustrated in Figure 4, LLM can assist digital twin systems with data generation, data alignment, and data fusion. Furthermore, LLM facilitates the development of learning strategies and the design of reward functions from textual data, enabling effective reinforcement learning. Moreover, LLM facilitates the embedding of physical knowledge and the design of loss functions, which are crucial for developing robust physics-informed machine learning models. The incorporation of minor and multimodal learning further augments the analytical capabilities, enabling DT to process diverse and complex datasets with minimal training. The synergy between LLM and DT establishes a robust framework for analysing real-time data and decision-making, as outlined in this section.

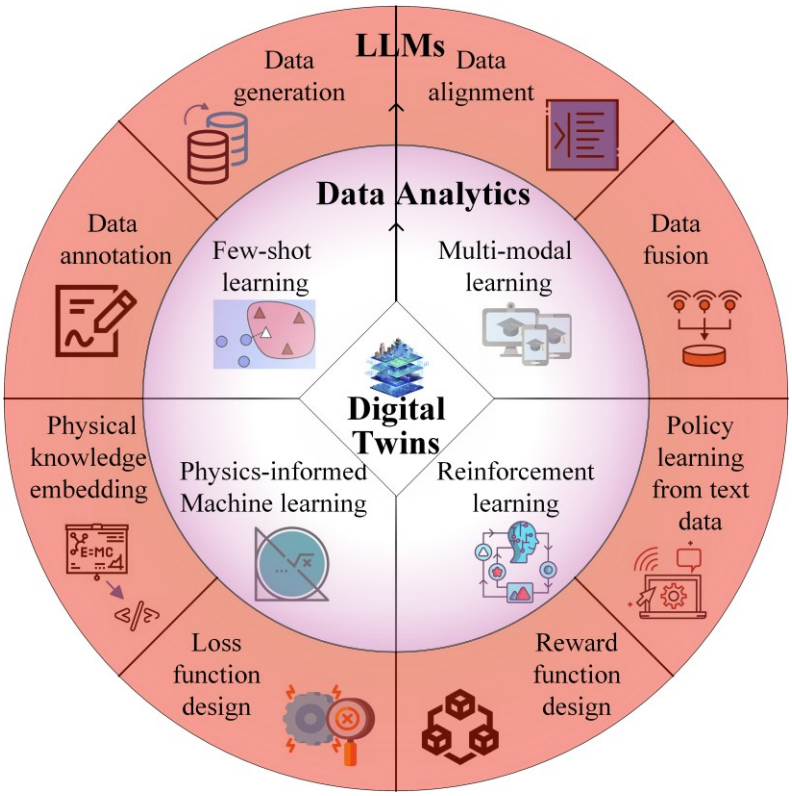


Figure 4. LLM for DT's advanced data analytics.

3.3.1 Reinforcement Learning for Dynamic Update of DT

DT needs to constantly reflect the state of the physical system and adjust accordingly, Reinforcement learning (RL) can help DT optimize its operation and maintenance strategies in real-time by continuously learning the feedback from the physical system. RL is a type of machine learning in which an agent learns to make decisions by performing actions in an environment to achieve maximum cumulative reward [95]. In DT, RL plays an important role. RL is not only an optimization tool but also provides DT with the ability to make dynamic decisions and adjust according. LLM can play an important role in enhancing reinforcement learning in several ways:

Policy learning: Policy learning is a fundamental aspect of reinforcement learning, involving the development of a policy by an agent to map states of an environment to actions to maximize cumulative reward [96]. In DT, policy learning continuously adapts to optimize operation, maintenance, and control policies in response to changes in the physical system and its surroundings. LLM can analyze extensive text data, such as manuals, operational logs, and research articles, to extract prevailing policies and strategies employed in analogous contexts. Through comprehension and summarization of documents, LLM lays the groundwork for formulating initial policies for DT agents, significantly reducing the time and effort needed to commence from scratch. LLM also enables interactive learning sessions between RL agents and human operators, during which the agents can pose queries, receive feedback, and modify their strategies in response. This interaction is especially beneficial in complex scenarios where human expertise is necessary to make sophisticated decisions.

Reward function design: Reward function is a critical component in RL, that defines the goals of the learning agent by quantifying the desirability of its actions [97]. The reward function provides feedback to the agent, guiding it toward behaviors that maximize cumulative rewards over time. The design of an effective reward function is essential for ensuring that the agent makes decisions that align with the desired outcomes. LLM can analyze textual data such as project documentation, user requirements, and industry standards to identify the objectives and constraints of DT. By comprehensively understanding goals, LLM can assist in the

decomposition of reward functions that focus on positively impacting the aim while still maintaining required conditions and boundaries [98]. A well-designed has a great impact on the performance and efficiency of the DT by promoting optimal behavior in various scenarios.

LLM also incrementally updates the reward function based on DT performance feedback. They adapt reward functions to changes in the operating environment, ensuring that agents remain effective over time. In Industry 5.0, DT needs to balance multiple aims such as efficiency, safety, and cost, and LLM can help design reward functions that appropriately weigh these objectives, ensuring that agents make balanced decisions that optimize the overall performance of the DT. In addition, LLM can generate detailed explanations of reward functions, making their design more transparent and ensuring alignment with DT learning objectives, thereby building trust and adoption.

3.3.2 Physics-informed Machine Learning for Enhanced DT Modeling

DT can simulate, monitor, and optimize real-world systems in industrial applications. These services rely on advanced modeling techniques to accurately reflect physical processes and make real-time decisions. Physics-informed machine learning (PIML), also known as knowledge-guided machine learning (KGML), is an approach that integrates the principles and laws of physics into machine learning models [99, 100]. PIML integrates physical laws into DT models, improving accuracy, generalizability, and interpretability. PIML incorporates underlying physical laws as prior knowledge, which is then utilized during the training process [101]. It ensures that the DT not only fits observed data but also respects the physical principles, resulting in more reliable models. In this context, LLM further augments DT by developing and refining PIML algorithms. LLM can encode physical laws and principles into the ML model by understanding and interpreting textual descriptions of physical laws [102]. This ability allows LLM to provide insights for the design of physics-informed network structures. They integrate this knowledge into the model, ensuring the model follows the relevant physical principles, improving the accuracy and robustness of predictions.

Loss function design is another essential component in PIML [103]. The main objective in

training an ML model is to reduce this loss function, thereby enhancing the precision and effectiveness of the model [104]. The effective loss function is crucial because it directly impacts the learning performance of the model. LLM can analyze requirements and generate loss functions designed for the objectives. LLM can propose new formulations that incorporate domain-specific knowledge and constraints, enhancing model performance in specialized tasks. LLM also can review and suggest improvements to existing loss functions based on the latest research and best practices to make loss functions more robust to noise and outliers, improving the overall robustness of the model. Furthermore, LLM can suggest optimal hyperparameter settings for the loss functions, such as weights for different terms in a composite loss function, and conduct hyperparameter tuning to ensure that the loss function effectively guides the model toward the desired outcomes.

3.3.3 Few-shot Learning for Rapid DT Adaptation

A key advantage of DT is the ability to learn from dynamic environments, which requires efficient data analysis and modeling techniques. Few-shot learning is a powerful approach to support DT by enabling rapid adaptation from minimal data, making DT highly flexible and efficient in diverse industrial applications. Few-shot learning aims to enable models to learn new tasks using only a small amount of training data [105]. Traditional ML models generally need substantial amounts of labeled data to achieve good performance, but few-shot learning techniques seek to overcome this limitation by leveraging prior knowledge and learning strategies to make accurate predictions from minimal data. It is useful in scenarios where labeled datasets are hard to obtain.

LLM plays a crucial role in enhancing the data analysis capabilities of DT by supporting few-shot learning in several ways. First, in data annotation, LLM can automatically generate labels for new datasets [106], which is a cost-efficient alternative to human-labeled data [105]. By understanding context and applying appropriate tags, LLM significantly reduces the manual effort required for data labeling. They ensure consistency and maintain high quality in annotations by following predefined guidelines and leveraging their extensive language understanding capabilities. In this way, DT can quickly adapt to new scenarios with limited

labeled data, thereby enhancing their efficiency and responsiveness.

In terms of data generation and augmentation, LLM improves the efficiency of DT for few-shot learning by creating synthetic data that closely resembles real-world data. It is useful for augmenting small datasets and improving model performance. LLM can simulate various scenarios and generate data for rare or difficult-to-obtain situations, which helps the model learn different behaviors. Moreover, LLM can generate domain-specific data that captures the nuances and characteristics of fields, ensuring that the few-shot learning model is well-tuned to the target application.

Another way that LLMs are used for small amounts of learning in DTs is knowledge transfer. LLM can embed prior knowledge from large text corpora into DTs, thus improving their ability to make accurate predictions from limited data. Their deep understanding of context allows DTs to infer relationships and patterns from a minimum number of examples, which improves generalization and adapts them to specific tasks.

Overall, few-shot learning enhances the adaptability of DT by enabling them to generalize effectively from small amounts of data, making DT highly practical for a wide range of applications. LLM plays an important role in enhancing few-shot learning by automating data annotation, generating synthetic data, embedding prior knowledge, and fine-tuning models for specific tasks. The capability of few-shot learning can reduce the dependency on large datasets and enable the creation of robust, high-performing models across diverse domains.

3.3.4 Multi-modal Learning for Comprehensive DT Insights

DT integrates diverse data sources to create a detailed and accurate representation of the physical system. Multi-modal learning plays a crucial role in DT by integrating and analyzing data from different modalities, such as text, images, and sensor data, to build robust models that enhance operational efficiency[44, 107]. Multi-modal learning enables the synthesizing of diverse data types to establish a more accurate and virtual representation of the physical system to provide insights for improved decision-making and operational efficiency.

Data alignment is an important step of multi-modal learning. Firstly, LLM can understand and match the semantic content across different data types[108]. For instance, they can connect textual descriptions to corresponding images or sensor readings to ensure proper data synchronization. Temporal alignment is another crucial aspect, where LLM aligns time-series data from different sources to ensure that events and measurements occurring at the same time are correctly correlated, which is essential for applications that require real-time monitoring and data analytics. Additionally, for applications involving spatial data, such as geographic information systems (GIS), LLM can align data points across different spatial scales and resolutions, ensuring coherent integration [109].

Data fusion is another essential aspect of multi-modal learning [110]. LLM facilitates integrating heterogeneous data by extracting relevant features and combining them into a unified representation [111]. It can help create models that leverage the strengths of each modality. LLM can also generate cross-modal representations that capture the relationships between different data types, which are essential for tasks requiring an in-depth understanding of interactions across modalities, including correlating sensor data with maintenance logs in a manufacturing context. In addition, LLM can use its understanding of natural language to pinpoint and emphasize the important data for each modal, thereby refining feature extraction to form more informative and efficient models.

3.4 LLM for Digital Twins' Application

The DT has revolutionized numerous industries by providing a virtual model of physical systems that can be used for monitoring, simulation, and optimization. DT finds applications in diverse fields such as product design, manufacturing, maintenance, and recycling, which is demonstrated in Figure 5. The applications of LLM for DT's application.. The next step in this technological synergy is the incorporation of LLM into the digital twin framework. LLM brings the power of NLP, cognitive computing, and advanced data analytics, enhancing the capabilities

of DT significantly. It can improve natural language interaction, real-time data processing, predictive maintenance, and automated documentation, among other benefits, which can pave the way for smarter and more responsive industrial systems.

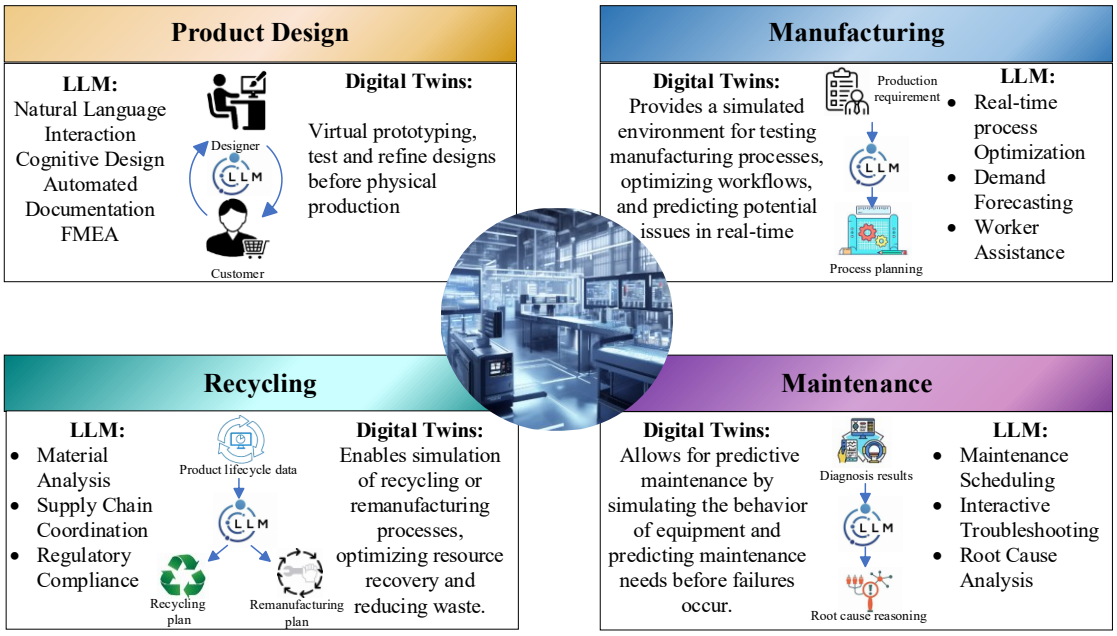


Figure 5. The applications of LLM for DT's application.

DT can be used to design new products in a more responsive, efficient, and informed manner [13]. DT offers transformative benefits in product design by providing a virtual replica of the product that can be tested and refined before physical production [111]. Virtual prototyping enables designers to simulate complex conditions, detect potential problems, and optimize the functionality of the design. In this stage, the integration of LLM with DT can enhance the aforementioned capabilities. First, the ability to NLP of LLM enables designers to interact with DT using commands and queries in natural language. For instance, a designer could ask, *"What would be the impact of increasing the material thickness by 10%?"*, and the LLM can interpret the query, retrieve relevant data, and provide a detailed simulation result or recommendation, which can simplify interaction and reduce the technical barrier for non-experts. Second, LLM can analyze design data and suggest improvements or alternatives based on historical data and trends. For example, based on the analysis of trends in material usage or failure records, the LLM can recommend a material that balances cost-efficiency with durability. Third, LLM can generate and update design documentation, including specifications, user manuals, and design

change logs. It ensures that documentation remains consistent and up-to-date, even as iterative changes occur during the design process. By reducing the workload in terms of communication, designers can focus more on innovation. Above all can help designers make decisions and lower the administrative burden, which leads product design more user-friendly and human-centered. Additionally, LLM can assist in conducting FMEA by identifying potential failure points and suggesting mitigation strategies based on past data [112] and predictive analytics. LLM can enhance virtual prototyping by interpreting simulation results and suggesting optimizations in real time [113]. It can improve product reliability and safety, reducing the risk of costly recalls and enhancing customer satisfaction, which also realizes Industry 5.0 in resilience and sustainability. LLM can process multi-modal inputs to support optimization, which includes textual and time-series data from the DT. For example, by considering constraints such as cost, environmental impact, and performance, the LLM can recommend design configurations aligned with Industry 5.0 goals of resilience and sustainability.

In the manufacturing industry, DT provides advanced virtual models of production systems and is an important tool for optimizing operations and improving efficiency. Virtual replicas can simulate production processes. The ability to provide detailed and in-depth analysis of production workflows, equipment performance, and resource utilization helps manufacturers identify bottlenecks, predict failures, and optimize processes [35]. The integration of LLM with DT improves decision-making and strategic planning capabilities through advanced data analytics and NLP. LLM enhances the predictive capabilities of DT by analyzing real-time data from DT to detect equipment conditions and predict potential failures. LLM enhances DT's predictive capabilities by analyzing real-time data from DT to detect equipment conditions and predict potential failures. For example, if a motor is vibrating abnormally, LLM can identify the root cause and recommend an appropriate maintenance program. In addition, LLM can identify inefficiencies and suggest adjustments to optimize production workflows. For example, based on sensor feedback, LLM suggests recalibrating equipment settings to reduce energy consumption or adjusting task sequencing to eliminate downtime. LLM also allows users to quickly access relevant information and get real-time help in dealing with tasks and problems. Worker productivity is increased, and training time is reduced. In addition, LLM utilizes

historical data, market trends and other external factors to make accurate predictions of future product demand. It helps companies adapt to a rapidly changing marketplace and is key to staying afloat.

Predictive maintenance is a proactive approach to equipment maintenance, and DT can increase productivity in predictive maintenance by identifying problems early and providing new perspectives continuously [115] through the analysis of sensor data. Furthermore, DT enables the virtual testing and refinement of maintenance strategies before their implementation in physical systems. LLM can further enhance these capabilities and minimize operational disruptions, enhancing equipment reliability, and reducing maintenance costs [112]. Specifically, LLM analyzes data, predicts faults, and gives recommendations for maintenance solutions, which can be simulated in DT before deployment is implemented. LLM guides maintenance personnel through diagnostic procedures, analyses root causes, and offers solutions based on data from DT. It can lead to more effective and long-lasting solutions. It also contributes to continuous improvement in maintenance practices. Additionally, LLM optimizes maintenance schedules by considering equipment usage patterns, production schedules, and historical maintenance data. In the context of fault self-healing, LLM assesses issues and determines whether they can be autonomously resolved. They provide instructions for recovery procedures and, in collaboration with the digital twin, simulate these procedures to validate their feasibility.

DT facilitates comprehensive simulation and analysis of diverse recycling strategies, optimizing resource recovery and reducing waste. It offers transformative potential in the field of recycling. DT enables the monitoring of materials throughout their entire lifecycle, from initial usage to recycling and remanufacturing [113, 114]. LLM can analyze data from DT to assess the composition and condition of materials to evaluate the recyclability of different materials, even identifying materials that present challenges in the recycling process [115]. LLM-DT can sort recyclable materials more accurately and efficiently, thereby increasing the quality and quantity of resources recovered. LLM can also simulate different recycling processes and predict the results to determine the most efficient and sustainable methods.LLM

can track changes in regulations and update compliance strategies accordingly, ensuring that recycling practices are in compliance with all legal and environmental standards. Additionally, LLM can automatically record and report on recycling, ensuring that reports are accurate, up-to-date, and facilitate decision-making and compliance. Finally, LLM can coordinate the supply chain to reduce transportation costs, minimize delays, and ensure a steady supply of recyclable materials to processing facilities[116].

4 Challenges and Opportunities

The integration of LLM with DT in Industry 5.0 presents various challenges and opportunities, especially in terms of illusion, bias, speed of reasoning, collaboration, and the overarching themes of human-centeredness, resilience, and sustainability. This section discusses these challenges, exploring how illusions in LLM can lead to inaccurate industrial decisions, the propagation of biases in DT, and the urgent need for fast speed of reasoning in industrial applications. In addition, it explores the role of agents in facilitating effective collaboration between DTs and LLMs and ensuring data transformation and interaction accuracy. Finally, it emphasizes the need to combine LLM-DT integration with human-centered, resilient, and sustainable industrial practices, laying the groundwork for a comprehensive discussion on how to mitigate challenges and leverage opportunities to improve industrial efficiency and sustainability.

4.1 Hallucination of LLM

Hallucinations in LLM refer to instances where the model generates outputs that are sound plausible but factually incorrect or nonsensical [52]. This poses a significant challenge in industrial applications, where inaccurate data or insights can lead to wrong decisions, jeopardizing operational efficiency, safety, and productivity. For instance, if LLM provides faulty recommendations for machinery maintenance based on hallucinated data, it could lead to unnecessary downtime, increased maintenance costs, or even catastrophic equipment failures. The complex nature and lack of transparency in LLM make identifying and rectifying these

errors difficult, further exacerbating the challenge, can guarantee the reliability and accuracy of information and insights provided by LLM is paramount, as any deviations from factual data can have widespread negative implications across various industrial applications.

Despite the challenges posed by hallucinations, there are several strategies to mitigate their impact and harness the power of LLM effectively. Designing a suitable prompt can significantly improve the accuracy of LLM prediction timing data [117]. The prompt should contain an introduction to the task context, and a description of the data characteristics (maximum value, minimum value, average value, etc.). Furthermore, similar to incorporating news knowledge in financial analysis [118], incorporating relevant information such as temperature, operation logs, etc. in industrial time-series analysis can further improve the accuracy. Another promising approach is integrating robust verification mechanisms within the DT framework. This involves cross-referencing LLM outputs with trusted data sources and leveraging DT simulations to test the feasibility and accuracy of LLM recommendations. By running LLM-generated insights through DT simulations, stakeholders can validate their plausibility, filtering out erroneous or hallucinated information. Continuous monitoring and feedback loops can further enhance this process by identifying and correcting hallucinations in real time. Additionally, one source of hallucinations is LLM's lack of relevant knowledge or internalized false knowledge [75], so domain-specific training can reduce the likelihood of hallucinations. The training datasets to the specific industrial context can make sure the models are aligned with the specific scenarios. The employment of hybrid models that combine rule-based systems with LLM can also provide an additional layer of accuracy, as rule-based systems can serve as a check against hallucinations [119]. KG is another important technique to lower the hallucinations. It can be used for RAG or provide triples for fine-tuning. These strategies collectively enhance the reliability of LLM in industrial applications, ensuring that the integration of LLM and DT delivers accurate and actionable insights.

While the above methods can improve the accuracy of LLM decision-making, incorrect inference is still inevitable. Hence, the reflection of the incorrect inference from LLMs needs to be implemented [120]. The inference of LLM can be evaluated by another evaluator LLM

and the evaluation results can be judged by an action LLM to generate suitable actions for DT. Memory modules are required to enable LLMs can learn from short-term and long-term experience, which can be helpful for the LLM to achieve consistent improvement. A potential self-reflective framework is shown in Figure 6. Such a reflective way can be helpful to the LLMs to learn effectively from previous mistakes, thereby improving the performance of DT.

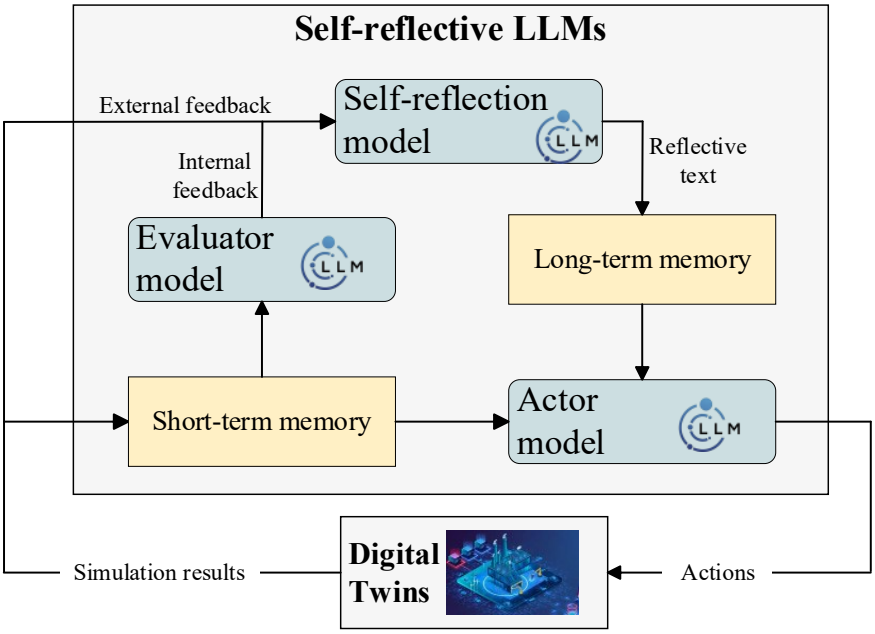


Figure 6 A Self-reflective Framework for Interactive-DT.

4.2 The bias of DT

The DT may inherit biases from the training database, the algorithms used, and the assumptions made during their development. These biases can lead to skewed insights, unfair outcomes, and systemic inefficiencies, which are particularly problematic in industrial settings where fairness and accuracy are crucial. To illustrate, the application of DT in manufacturing may result in the preferential selection of certain production methods due to the influence of biased input data. This can lead to inconsistencies in product quality and a reduction in operational efficiency. In the context of predictive maintenance, the application of biased DT may result in the prioritization of maintenance on specific equipment, based on historical data, while other critical equipment may be overlooked, potentially leading to unexpected failures and costly

periods of downtime. Furthermore, the introduction of biased DTs in workforce management can give rise to unfair labor practices, which may have an adverse impact on worker morale and productivity. It is therefore imperative to address these biases in order to ensure that DT provides accurate and impartial insights, which can subsequently enhance overall operational performance.

LLM has significant potential for the identification and reduction of DT bias. One effective approach is to utilize LLM for comprehensive bias detection and analysis. LLM is capable of examining extensive datasets to identify anomalies that suggest the presence of bias, thereby providing insight into potential areas of bias in DTs. By cross-referencing with disparate representative datasets, LLM can assist in guaranteeing that the data entered into the DT is balanced and comprehensive, thereby reducing the probability of biased outcomes. Moreover, LLM can assist in the creation of algorithms that are more inclusive and unbiased by employing fairness-aware training techniques. These techniques include adapting the training process to account for and correct for bias, thereby creating fairer models. Additionally, LLM improves the transparency and explainability of DT's internal decision-making processes. By explaining the rationale behind its recommendations, it enables stakeholders to understand and scrutinize the factors that influence outcomes, thereby reducing the impact of bias. Regular audits using LLM further enhance bias detection and correction, thus contributing to the continuous improvement of the DT system.

4.3 The Inference Speed of LLM

Inference speed is a critical factor in the practical deployment of DT in industrial applications. Effective digital twin modeling relies solely on the coherent analysis and rapid interpretation of multiple relevant attributes of a physical object or process [121]. LLM is also computationally resource intensive. Slow inference times affect the effectiveness of real-time monitoring, control, and optimization of industrial processes. In manufacturing, delayed reasoning may lead to slower responses to machine anomalies, which reduces efficiency and

increases the risk of equipment failure[122]. In supply chain management, sluggish inference speeds can delay decision-making processes, leading to inefficiencies in logistics and inventory management [82].

There are a number of ways to improve the speed of inference during the integration of LLM with DT. One approach is to optimize the LLM architecture to balance accuracy and speed. Methods such as model pruning, quantization, and distillation can help reduce the computational burden on the LLM, speeding up inference without significantly degrading performance. [122-124]. Additionally, leveraging specialized hardware such as GPUs and TPUs can accelerate inference times, which enables real-time processing [125]. Deploying DT on edge computing platforms also can reduce the burden on the cloud and increase inference speed, thereby reducing latency [125-127]. Furthermore, LLM facilitates dynamic resource management by allocating computing resources based on real-time demand. By combining parallel processing and asynchronous algorithms, LLM can process multiple tasks simultaneously, further speeding up the inference process. Integrating these strategies ensures that the combination of LLM and DT provides fast, reliable, and responsive performance to meet the stringent requirements of industrial applications.

4.4 Agent for the collaboration of DT and LLM

LLM-DT faces several challenges in data collaboration. DT generates complex outputs, including real-time sensor data, simulation results, and predictive analytics, while LLM produces extensive textual outputs, offering insights, recommendations, and analysis. Effective collaboration between the two requires distilling DT-generated data into concise, actionable instructions for the LLM. Conversely, the lengthy text produced by the LLM needs to be transformed into a structured data format suitable for DT input. Therefore, key information must be extracted from LLM responses and appropriately structured to ensure compliance with DT's data requirements. As shown in Figure 7, integrating the DT and LLM through an intermediary agent that enables interaction between them can ensure the accuracy and relevance

of the translation.

By employing advanced data processing algorithms, agents can parse the DT's comprehensive data streams, extract relevant information, and convert it into a format that LLM can effectively utilize to provide LLM with the most relevant and contextualized data. Conversely, the agent can distill the large amount of textual information provided by the LLM to identify key insights and recommendations. The distilled information is then converted into a structured data format that meets the DT input requirements.

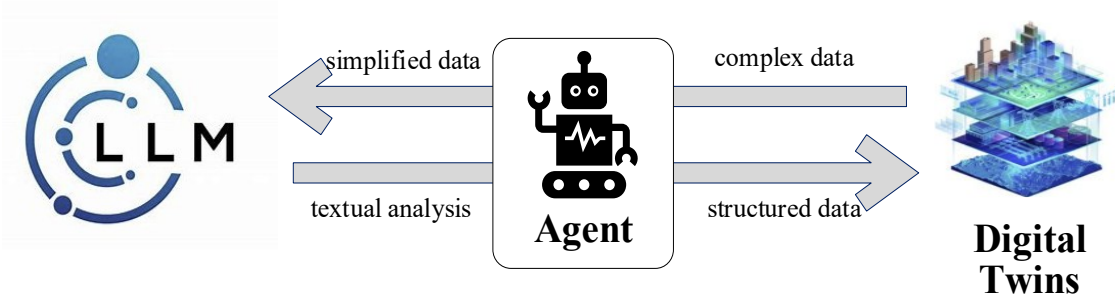


Figure 7 An agent for the collaboration of DT and LLM.

4.5 Human-centricity, resilience, and sustainability-oriented LLM-DT

Industry 5.0 emphasizes human-centricity, resilience, and sustainability [2, 3, 128], Interactive-DT presents an opportunity to enhance these core features. Firstly, the complexity of managing data generated by DTs, which also includes the challenge of ensuring the privacy and security of the data, necessitates that LLMs possess the capacity to process and interpret this information in real-time. Secondly, the evolving nature of industrial environments necessitates the development of adaptive LLM algorithms capable of continuous learning and evolution, which represents a significant technical challenge. Furthermore, the integration of LLM with DT must be aligned with sustainable development goals, which include optimizing resource utilization and minimizing environmental impact. This necessitates the implementation of advanced energy-efficient computational methods. Additionally, the "human-centered" aspect presents the challenge of developing interfaces that are intuitive and easily comprehensible for non-technical users. Finally, the resilience aspect requires that the LLM-DT system be capable of

withstanding failures and disruptions. This requires the incorporation of sophisticated fault-tolerance mechanisms and comprehensive disaster recovery plans.

The above challenges open up several potential research directions. First, LLMs with privacy-preserving techniques can be further explored, which can prevent privacy data leakage. Another direction is to explore new LLM architectures that are more energy efficient and scalable to achieve sustainability goals, which may involve research in quantum computing or neuromorphic computing, which promise significant advances in computational efficiency. In addition, future research could also focus on enhancing the user experience by developing more intuitive and interactive interfaces that allow users to use the LLM-DT system effectively. This could involve the use of natural language processing and augmented reality technologies to create a more immersive and user-friendly environment. Finally, it is critical to investigate the recovery mechanisms of the LLM-DT system, including the development of robust fault detection and recovery strategies to ensure continuous operation even in the event of an outage.

5 Conclusion

Integrating LLM and DT in Industry 5.0 can enhance all aspects of industrial operations and enable smarter, more efficient, and adaptive manufacturing environments. This paper first reviewed the latest research on LLM and DT. Then, a framework named Interactive-DT for integrating LLM and DT in Industry 5.0 is presented. By examining the role of LLM in advanced data analytics, product design, manufacturing, maintenance, and recycling, the research demonstrates how LLM-DT can improve efficiency, accuracy, and innovation. In addition, this paper explores the co-computing issues that may be encountered when deploying LLM-DT in practice. Finally, mitigation strategies are proposed for challenges such as illusion, bias, and inference speed. Agents are also used to enable closer collaboration between LLM and DT. This study contributes to the understanding of how LLM can enhance DT capabilities. Drawing from the aforementioned discussion, this paper offers perspectives on the research and application of the amalgamation of LLM and DT, which are poised to deliver substantial advantages to industrial operations.

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CRediT authorship contribution statement

Chong Chen: Methodology, Investigation, Conceptualization, Writing – original draft. **Kuanhong Zhao:** Writing –original draft, Methodology, Investigation, Data curation Conceptualization. **Jiewu Leng:** Writing – review & editing, Methodology, Conceptualization, Supervision. **Chao Liu:** Writing – review & editing, Formal analysis, Conceptualization. **Junming Fan:** Writing – review & editing, Investigation, Conceptualization, Validation. **Pai Zheng:** Writing – review & editing, Investigation, Supervision.

Data availability

Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

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